# JavaScript Assignment - Objects, Array of Objects & Higher Order Functions

## Assignment 1: Student Performance Analytics System

Problem Statement:
A university wants to analyze student performance using JavaScript Objects and Higher Order Functions.

Input:
const students = [
{id:1,name:'Rahul',course:'MERN',marks:85},
{id:2,name:'Priya',course:'Java',marks:92},
{id:3,name:'Aman',course:'Python',marks:65},
{id:4,name:'Neha',course:'MERN',marks:78},
{id:5,name:'Rohit',course:'Java',marks:95}
];

Tasks:
1. Create an array containing only student names using map().
2. Find students scoring above 80 using filter().
3. Calculate total marks using reduce().
4. Calculate average marks.
5. Find the topper.
6. Sort students by marks in descending order.

Expected Output:
Names: ['Rahul','Priya','Aman','Neha','Rohit']
Total Marks: 415
Average Marks: 83
Topper: Rohit (95)

## Assignment 2: Amazon Product Dashboard

Problem Statement:
Manage product information for an e-commerce platform.

Input:

```
const products = [
{id:1,name:'Laptop',price:70000,stock:10},
{id:2,name:'Phone',price:25000,stock:15},
{id:3,name:'Headphone',price:3000,stock:20},
{id:4,name:'Smart Watch',price:5000,stock:12}
];
```

Tasks:
1. Display all product names.
2. Find products costing above 10000.
3. Calculate inventory value (price × stock).
4. Find most expensive product.
5. Apply 10% discount using map().
6. Sort products by price descending.

Expected Output:
Most Expensive Product: Laptop

## Assignment 3: Employee Payroll Management System

Problem Statement:
A company wants to analyze employee salary data.

Input:
```
const employees = [
{id:101,name:'Amit',salary:35000,department:'HR'},
{id:102,name:'Riya',salary:50000,department:'IT'},
{id:103,name:'Karan',salary:45000,department:'IT'},
{id:104,name:'Pooja',salary:60000,department:'Finance'}
];
```

Tasks:
1. Display all employee names.
2. Find employees earning above 45000.
3. Calculate total salary expense.
4. Find highest-paid employee.
5. Count employees department-wise.
6. Sort employees by salary.

Expected Output:
Highest Paid Employee: Pooja

Total Salary Expense: 190000

## Assignment 4: Online Order Processing System

Problem Statement:
An online shopping website stores customer orders.

Input:
```
const orders = [
{id:1,customer:'Rahul',amount:1200},
{id:2,customer:'Priya',amount:800},
{id:3,customer:'Aman',amount:2500},
{id:4,customer:'Neha',amount:1500}
];
```

Tasks:
1. Find orders above Rs.1000.
2. Calculate total revenue.
3. Calculate average order value.
4. Apply 5% discount to every order.
5. Sort orders by amount.

Expected Output:
Total Revenue: 6000
Average Order Value: 1500

## Assignment 5: University Management System (Comprehensive)

Problem Statement:
This assignment combines Objects, Arrays, Functions and Higher Order Functions.

Input:
```
const students = [
{id:1,name:'Rahul',course:'MERN',fees:45000,attendance:85},
{id:2,name:'Priya',course:'Java',fees:50000,attendance:92},
{id:3,name:'Aman',course:'Python',fees:35000,attendance:70},
{id:4,name:'Neha',course:'MERN',fees:45000,attendance:95},
{id:5,name:'Rohit',course:'Java',fees:50000,attendance:88}
];
```

Tasks:
1. Display all student names.
2. Find attendance above 80.
3. Calculate total fees collected.
4. Calculate average attendance.
5. Find highest attendance student.
6. Count students course-wise.
7. Search student by name.
8. Create array of names using map().
9. Sort students by attendance.
10. Generate complete report.

Expected Output:
Highest Attendance: Neha (95)
Total Fees Collected: 225000
MERN:2
Java:2
Python:1